

Known at the origin  $o$ 

Three orthogonal blocks

Error-weighted handover

Output

**Unit's own history**

hourly load, temperature  
and calendar before  $o$



pseudo-origins  $o - 744k$



completed forecasts give every  
candidate its own error record

**TimesFM 2.5**

zero-shot, load only  
1,344 h  $\rightarrow$  744 h, issued at  
 $o$  and at  $o - 744k$  ( $k \leq 6$ )

**Level  $\ell$** 

six historical candidates: recent, long-history  
and annual means, with and without weather

+ TimesFM window mean

weights  $\propto 1/\text{MSE}$  at pseudo-origins, shrunk to equal ( $K_0 = 8$ )

**Centred daily path  $b$** 

seven historical candidate paths: 8-, 4- and 2-week  
day types, the annual window, the weather response

same error weighting ( $K_0 = 2$ );  $\sum_d b_d = 0$

**Within-day shape  $w$** 

day-demeaned TimesFM trajectory  
the foundation model's most reliable block

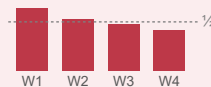
**Week-by-week  
handover**

$$m_d = (1 - \alpha_w) m_d^H + \alpha_w m_d^T$$

$m^H$ : history ( $\ell + b$ )

$m^T$ : TimesFM daily means

$\alpha_w$ : least squares on  $\leq 6$   
pseudo-origin pairs (fixed  
division vs TimesFM),  
shrunk to  $1/2$  ( $K_0 = 2$ )

**ANKYRA**

$$F_{d,h} = m_d + w_{d,h}$$

projected onto  $F \geq 0$

no training on the  
target series

**off-state switch**

last 168 h  $\leq 10^{-6}$  kW  
 $\rightarrow$  TimesFM unchanged

**Readouts**

energy 744  $\ell$

peak  $\max_d (\hat{L}_d + A_{T_d})$

interval pseudo-origin  
residual quantiles

Exact identity:  $\text{MSE} = \ell(e)^2 + \langle b_d(e)^2 \rangle_d + \langle w_{d,h}(e)^2 \rangle_{d,h}$  — the blocks are orthogonal, so each source can be scored and replaced block by block.

from the unit's own history    from TimesFM 2.5    ANKYRA's handover